

**Green Transition Alpha Fund:
A Critical Review of a Benchmark-Aware Climate
Transition Strategy**

Individual Project Write-Up

DSS 5203 ESG Data for Sustainable Finance and Investments

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Table 1. Summary of Green Transition Alpha Fund design

Fund type	Active, long-only global equity fund
Benchmark	MSCI ACWI
Core theme	Climate transition leaders reducing carbon intensity and/or investing in clean technologies
Screening logic	Negative exclusions + minimum ESG threshold + forward-looking transition score
Portfolio shape	40-60 names; benchmark-aware construction; capped individual stock weights
Key portfolio metrics	WACI, weighted-average ESG score, controversy signals, issuer thesis checks, stewardship activity

This individual write-up builds on our group presentation for the Green Transition Alpha Fund, an active, long-only global equity strategy focused on climate transition leaders. Rather than restating the pitch, I develop four issues that needed more space in class: how the product should be positioned, why a transition score matters more than a static ESG rating, why lifecycle governance is central to credibility, and what the backtesting evidence actually shows. My view is that the fund is viable, but only if it is presented honestly as a benchmark-aware transition product with explicit trade-offs, not as a simple shortcut to 'green alpha' (Avalokite, 2026; Khew, 2026f).

1. Why the climate transition can support an equity strategy

The group project starts from a reasonable investment premise: decarbonisation is not only an environmental objective but also a long-duration economic reallocation. Carbon pricing now covers around 28% of global greenhouse gas emissions, creating a growing cost differential between firms that can operate under tighter climate policy and those that remain highly exposed to fossil-intensive business models (World Bank, 2025). At the same time, capital spending is moving toward low-emissions technologies. The International Energy Agency estimates that energy investment will rise to about USD 3.3 trillion in 2025, with roughly USD 2.2 trillion going to clean energy technologies and infrastructure - about twice the amount going to oil, natural gas,

and coal (IEA, 2025). In other words, policy, capital expenditure, and end-market demand are increasingly pointing in the same direction.

That said, not every climate-related fund is the same. One important point that the presentation could only mention briefly is product classification. In the course framework, ESG-promoted, thematic, and impact funds are different product types with different expectations around objectives, data use, and outcomes (Khew, 2026b). In my judgement, our design fits best as a benchmark-aware thematic or ESG-promoted strategy, rather than a pure impact fund. The portfolio is still constructed in public equities, benchmarked against MSCI ACWI, and justified partly by risk-adjusted return. That makes it more disciplined and realistic, but it also means the strategy should not overclaim direct climate impact.

This classification issue also affects external credibility. Regulators have tightened expectations around sustainability-related fund names and disclosures. MAS has set disclosure and reporting expectations for retail ESG funds in Singapore, while ESMA's guidelines require stricter alignment between a fund's name and its actual holdings in Europe (MAS, 2022; ESMA, 2024). Even though our proposed product is not a retail vehicle and does not use "impact" in its name, the lesson is the same: the manager must define clearly what "transition leader" means, what share of the portfolio must meet that definition, and what metrics will be reported to clients. In my view, the fund becomes stronger when it avoids overstating what it is. It is not an impact-first solution. It is a benchmark-aware climate transition equity strategy that uses ESG and climate data to pursue alpha while lowering carbon exposure.

2. From static ESG ratings to a dynamic transition score

A second idea worth developing is the difference between using ESG ratings as a broad quality filter and using transition data as the real engine of stock selection. Our group design uses a five-stage funnel: global investable universe, negative screens, a minimum ESG rating, transition scoring, and portfolio construction. That is already a more credible framework than treating ESG labels as sufficient evidence of climate relevance, because it links security selection to forward-looking transition characteristics rather than backward-looking reputation measures alone (Avalokite, 2026; Khew, 2026f).

The crucial point is that static ESG ratings alone are not enough for a climate-transition fund. MSCI's methodology is industry-relative, which is useful for peer comparison, but it also means that a company can look strong on a relative basis without necessarily being a compelling

transition leader in absolute terms (MSCI, 2024). That is why our project appropriately adds more forward-looking indicators: green revenue share, low-carbon capital expenditure, Science Based Targets initiative (SBTi) alignment, and clean-technology patents (Avalokite, 2026). Course materials make a similar distinction between backward-looking data, future-based targets, and forward-looking signals that indicate where a company is heading rather than only where it stands today (Khew, 2026a; Khew, 2026f). For a transition strategy, that distinction is fundamental.

I would go one step further in my own interpretation of the project. A strong transition fund should not invest only in already-green companies. It should combine “transition enablers” and “transition improvers.” Enablers are firms whose products directly support decarbonisation, such as electrical equipment, renewable technology or efficiency infrastructure. Improvers are high-emitting or hard-to-abate firms that have credible decarbonisation pathways, rising green capex and genuine operational transition potential. The course lecture on climate-transition approaches explicitly highlights the importance of looking beyond pure green sectors and considering brown-to-green opportunities as well (Khew, 2026f). Adding more of these improvers could make the fund more robust, because it would reduce dependence on crowded green-growth segments that are highly sensitive to interest rates and momentum reversals.

This point also connects to one of the project’s stated risks: ratings dispersion. The same company can look different across ESG providers because methodologies vary in scope, weighting, and treatment of controversies. That is not a minor technical issue. It can materially alter the investable universe and create false precision in portfolio design (Berg et al., 2022).

3. Lifecycle management is the real credibility test

In my view, the strongest part of our group project is not the opening investment story but the investment lifecycle design. Course materials repeatedly emphasise that ESG integration should run through the whole investment cycle: screening, integration, monitoring, engagement, proxy voting, and reporting (Khew, 2026c; Khew, 2026d). Our slide on lifecycle management reflects that logic. After the investable universe is defined, the fund conducts deeper analysis on transition readiness, carbon-cost sensitivity, policy exposure, and management credibility; once invested, the portfolio is monitored using WACI, weighted-average ESG score, controversy signals, and issuer-level thesis checks; these signals then feed into stewardship and reporting (Liu, 2026).

This matters because ESG credibility breaks down when a manager treats sustainability as a one-off entry filter. A climate-transition strategy should continue testing whether each holding is

still consistent with the thesis. Has capex remained aligned with transition claims? Are emissions intensity and disclosure quality improving? Have controversies changed the risk profile? Has management weakened its commitment, or is the market now paying too much for the same transition story? These are active portfolio questions, not static labeling exercises.

At the same time, the monitoring framework in our project should be interpreted with some caution. WACI is useful because it gives an intuitive benchmark-relative measure of carbon intensity, but it is not a full measure of real-world contribution. A fund can lower WACI simply by tilting away from carbon-heavy sectors, even if the companies it owns are not materially changing the real economy. The course slides make this clear by listing WACI, controversy scores, engagement counts and weighted-average third-party ESG scores as portfolio indicators that help describe a fund's ESG characteristics, not as complete proof of impact (Khew, 2026e). This is why I would pair WACI with at least three additional disclosures: first, sector attribution, so clients can see whether lower carbon intensity comes mainly from selection or from sector underweights; second, a transition-plan metric, such as the share of holdings with validated science-based targets or rising green capex; and third, engagement outcomes, not merely engagement activity.

This distinction is important because our group conclusion uses the phrase “measurable, real-world impact.” I would use that wording more carefully. A public-equity fund can credibly claim lower financed-emissions intensity, clearer stewardship activity, and better exposure to climate-transition opportunities than a broad-market benchmark. What it cannot automatically claim is additional decarbonisation in the real economy. That depends on whether engagement changes behaviour, whether capital allocation affects cost of capital, and whether the manager can demonstrate influence beyond portfolio relabelling.

4. Which clients fit the product - and which do not

The target-client section of the presentation identifies pension funds, endowments and foundations, family offices, and sovereign wealth funds (Avalokite, 2026; Liu, 2026). I broadly agree with that framing, but I would narrow it further after looking at the backtesting results. Pension funds are indeed a natural fit because many now have net-zero commitments and reporting obligations. Endowments and foundations may also fit if they want a climate-tilted equity sleeve that still remains benchmark-aware. By contrast, I would be more cautious about marketing the product to family offices unless they explicitly accept thematic volatility.

However, the backtest suggests that this should not be sold as a core allocation for all institutions. The strategy appears better suited either as a satellite mandate within a larger diversified equity book or as a specialist thematic sleeve for clients that explicitly want transition exposure. That conclusion follows directly from the simulated risk profile. The management fee of 0.65% plus a 10% performance fee over MSCI ACWI with a 1% hurdle is commercially understandable for an active specialist fund, but it is only defensible if the manager can show genuine skill in stock selection and risk control (Avalokite, 2026). Otherwise, a lower-cost low-carbon index product may be more appropriate for many asset owners.

Table 2. Selected backtesting results (2019-2026)

Period	Fund NAV	Benchmark NAV	Interpretation
2019-2021	1.05 -> 3.90	1.00 -> 1.60	Fund strongly outperformed during the green-growth rally.
2022-2025	3.90 -> 1.45	1.60 -> 1.80	Large drawdown exposed rate sensitivity and concentration risk.
2025-2026	1.45 -> 2.15	1.80 -> 2.50	Fund rebounded faster, but still ended below benchmark.
Full period	11.7% ann.; Sharpe 0.3; MDD -66.7%; CI 65	14.2% ann.; Sharpe 0.7; MDD -33.5%; CI 168	Cleaner portfolio, but weaker full-cycle risk-adjusted performance.

Note. CI = carbon intensity (tCO₂e per USD million revenue).

5. What the backtest really tells us

Before interpreting the results, one clarification matters. After comparing the uploaded written backtesting note with the charts embedded in our presentation slides, I do not think they should be treated as one perfectly consistent backtest. The written note reports a rule-based 2019-2026 simulation with its own return, Sharpe ratio, maximum drawdown, and carbon-intensity results. By contrast, the slide visuals present a separate proxy analysis and are better read as supporting illustrations. For a classroom project this is understandable, but it means the performance evidence

should be read with some caution. In this essay, I therefore rely mainly on the written note for the formal interpretation, while using the uploaded slide figures to show the trade-off between transition exposure, volatility, and benchmark-relative performance. This inconsistency was not highlighted in the group presentation, but it is an important caveat for interpreting the strategy's performance claims.

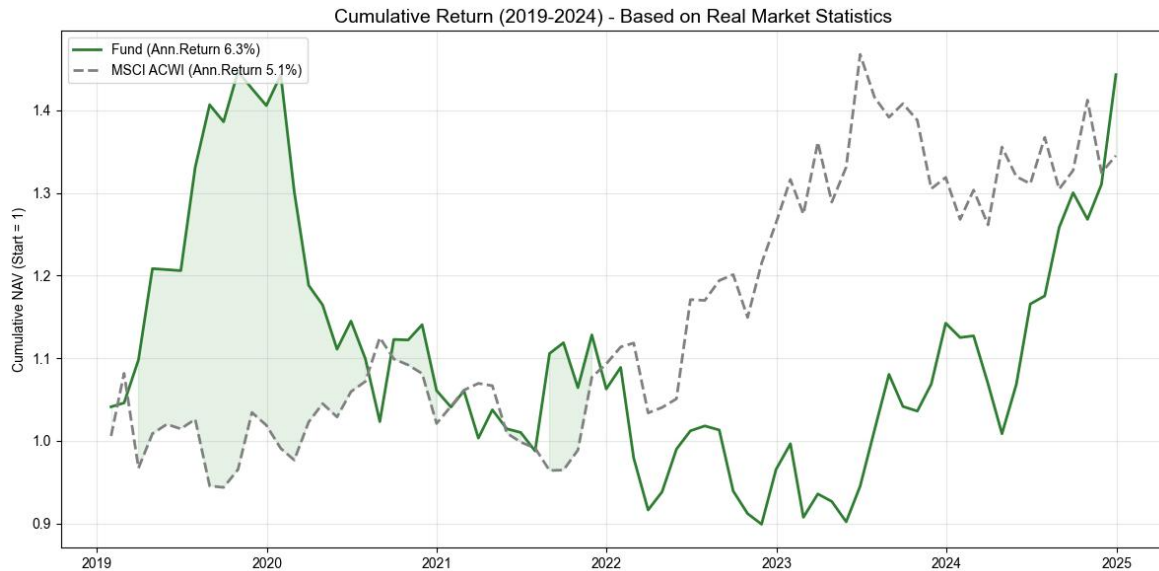


Figure 1. Cumulative return chart reproduced directly from Avalokite_PPT(1).pptx (Slide 11). These figures are presented for illustrative purposes only, as they are based on a separate proxy comparison using ICLN and MSCI ACWI.

The most useful reality check in the project is the backtesting exercise. In the early phase, the simulated fund looked exceptional: from 2019 to 2021, cumulative NAV rose from 1.05 to 3.90, compared with 1.00 to 1.60 for the benchmark. That is a 271% cumulative return for the fund versus 60% for the benchmark (Data Backtesting Team, 2026a). But the next phase is just as revealing. From 2022 to 2025, the fund fell from 3.90 to 1.45, a 63% peak-to-trough decline, while the benchmark rose from 1.60 to 1.80. Even after a strong rebound in 2025 to 2026, when the fund recovered 48% from 1.45 to 2.15, it still finished below the benchmark's 2.50 (Data Backtesting Team, 2026a).

The full-period metrics reinforce the same lesson. The fund was cleaner - carbon intensity of 65 tCO₂e per million dollars of revenue versus 168 for the benchmark, or about 61% lower - but it did not deliver better risk-adjusted performance. Annualised return was 11.7% for the fund versus 14.2% for the benchmark, while the Sharpe ratio was 0.3 versus 0.7 and maximum drawdown was

-66.7% versus -33.5% (Data Backtesting Team, 2026a). This is a useful result academically because it separates climate alignment from portfolio efficiency. Lower portfolio carbon intensity does not, by itself, prove better investment construction.

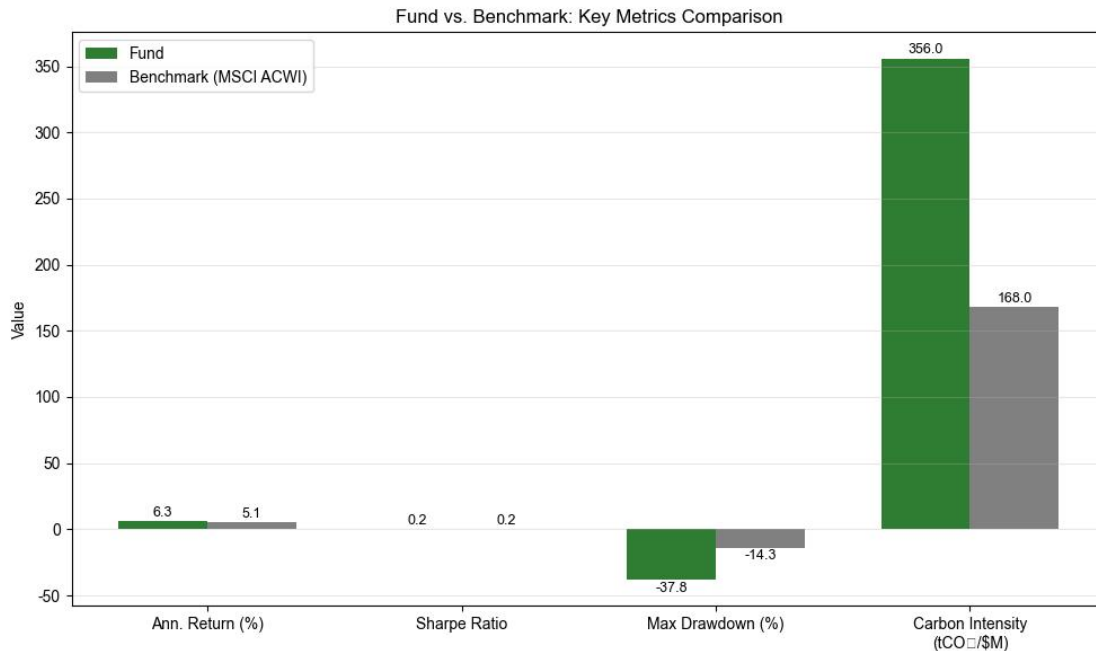


Figure 2. Key metrics chart reproduced directly from Avalokite_PPT(1).pptx (Slide 12). These figures are presented for illustrative purposes only, as they are based on a separate proxy comparison using ICLN and MSCI ACWI.

I do not think this invalidates the climate-transition thesis. What it shows is that implementation dominates narrative. A fund can be directionally right about the transition and still be a poor product if it expresses that thesis through an overly concentrated basket of high-beta green growth stocks. The backtest notes themselves reach a similar conclusion, pointing to weak valuation discipline and insufficient diversification (Data Backtesting Team, 2026a). That is why I would interpret the simulation not as evidence against transition investing, but as evidence against simplistic transition investing.

If I were refining the strategy after this exercise, I would make four changes. First, I would introduce stronger sector bands relative to MSCI ACWI to control unintended factor and industry bets. Second, I would balance transition enablers with credible improvers to reduce dependence on a single part of the value chain. Third, I would add an explicit valuation overlay so that the fund does not systematically overpay for green narratives. Fourth, I would define the stewardship

escalation path more clearly, including milestones for engagement, voting, and possible divestment when transition claims are not matched by execution.

Conclusion

The Green Transition Alpha Fund is a persuasive product idea because it is built around a structural economic transition rather than a vague sustainability label. The group presentation successfully established the headline logic: climate policy, capital spending, and client demand can create an investable tailwind for transition leaders. My own interpretation is slightly more cautious. The strategy is strongest when it is presented not as proof that ESG automatically outperforms, but as a benchmark-aware thematic product whose credibility depends on disciplined data use, realistic client positioning, and consistent lifecycle governance. It is one that offers a well-governed, data-disciplined way to access climate-transition exposure while acknowledging the limits of both ESG metrics and market timing.

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